

MAHMOUD ELLOUMI

+216 23 222 334 | mahmoudelloumi@gmail.com | Sfax, Tunisia
linkedin.com/in/mahmoud-elloumi-23222334- | mahmoud-elloumi.vercel.app

PROFILE

Computer Science and Applied Mathematics engineering student at ENIS. Across 2 internships and several projects, I have integrated LLMs and generative AI into full-stack applications using Ollama, Mistral AI and n8n. Seeking a **final-year internship (PFE)** to design **AI agents** that solve real business needs.

EXPERIENCE

- Full-Stack & AI Developer Intern – IDSOFT (ID Software Solutions)** Sfax | 07/2026 – 08/2026
AI-enhanced after-sales service (SAV) management app *React, Spring Boot, PostgreSQL, Ollama*
 - Built an after-sales web app (7-state service-request lifecycle) for 3 roles: customer, technician, manager.
 - Integrated 6 local generative AI features (Ollama, no data sent to third parties): troubleshooting assistant, natural-language data assistant, request classification, best-technician suggestion, report-writing assistance.
 - Designed a layered Spring Boot REST API (Swagger), secured with JWT, TOTP, BCrypt and role-based access control.
 - Created the manager dashboard: interventions by status and priority, revenue, employee performance.
- Mobile Developer Intern – LOGICOM Informatique** Sfax | 07/2025 – 08/2025
Product scanning mobile app *Flutter*
 - Built a Flutter QR-code scanning app covering 40 products, showing available prices and product details.
 - Delivered 3 screens (home, scanner, cart) connected to the company database.

PROJECTS

- SPARKY – Intelligent Business Document Processing (PFA)** 05/2026
Team of 3, Scrum (9 sprints) *Angular, FastAPI, Mistral AI, PostgreSQL, n8n*
 - Designed the 11-node n8n workflow (Google Sheets export, email notification) and built the Angular 20 frontend: scanner, price comparison, dashboard.
 - AI extraction of invoices, quotes and purchase orders with Mistral AI: 3–8 s per document vs 120–180 s for our n8n + Ollama prototype (90% faster).
 - Compared 3 suppliers across 10 items in under 1 minute, vs about 20 minutes by hand.
 - Containerized all services with Docker Compose; local fallback extraction (pdfplumber + regex) when the AI fails.
- Intelligent Traffic Monitoring – individual course project** 2025
 - Real-time vehicle detection and classification on video streams (YOLO, OpenCV, Python), with a dashboard counting vehicles by type.
- TERRANAUT – Drone-Based Detection of Sick Animals – team course project** 2024
 - Applied computer vision and deep learning (object detection) to thermal images captured by drones.

TECHNICAL SKILLS

AI & LLMs	LLMs (Ollama, Mistral AI, Qwen), Generative AI, AI agents, Machine Learning
Vision & Automation	Deep Learning, OpenCV, YOLO, PyTorch, OCR, n8n
Languages	Python, Java, JavaScript/TypeScript, C
Web & Mobile	React, Angular, Spring Boot, FastAPI, Node.js, Flutter
Data & DevOps	PostgreSQL, MySQL, MongoDB, Kafka, Docker, Git, Linux, Scrum

EDUCATION

Engineering Degree in Computer Science and Applied Mathematics – ENIS, Sfax	2024 – Present
Preparatory Cycle in Technology – IPEIS, Sfax (<i>top 100 in the national ranking</i>)	2022 – 2024
Baccalaureate in Technical Sciences – 15 November 1955 High School, Sfax	2022

CERTIFICATIONS

NVIDIA Getting Started with Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

VOLUNTEERING (IEEE)

Photographer – IEEE ENIS SB (2026) • Treasurer & Webmaster – IEEE Cyber Security ENIS (2025)
Social Media Manager & Photographer – TSYP 13 • Organizer – Pitch Yourself for a Career

LANGUAGES & INTERESTS

Arabic: native French: upper-intermediate English: intermediate Spanish: elementary
Interests: photography, art, sports, innovation